ERP Selection Executive Summary

MULTI-CRITERIA DECISION-MAKING (TOPSIS METHOD)

Project: ERP Vendor Evaluation
Model: 6 Vendors / 9 Criteria
Date: October 2026

> **Executive Overview:** This report provides a decision framework for evaluating six Enterprise Resource Planning (ERP) vendors across 9 quantitative and qualitative criteria (3 Cost, 6 Benefit parameters). Using the *TOPSIS* methodology, multi-objective metrics were synthesized into a single relative closeness index (C_i^*) to select the optimal platform.

TOP RANKED VENDOR
ERP 5
Closeness Score: 0.6558

RUNNER-UP VENDOR
ERP 6
Closeness Score: 0.5690

LOWEST MATCH
ERP 3
Closeness Score: 0.3539

1. Vendor Rankings & TOPSIS Closeness Scores

RANK	VENDOR	S_i^* (IDEAL)	S_i^- (ANTI)	C_i^* SCORE
1	ERP 5	**0.0225**	**0.0429**	**0.6558**
2	ERP 6	0.0282	0.0372	**0.5690**
3	ERP 4	0.0320	0.0342	**0.5167**
4	ERP 2	0.0303	0.0323	**0.5161**
5	ERP 1	0.0367	0.0292	**0.4434**
6	ERP 3	0.0424	0.0232	**0.3539**

Relative Closeness Index (C_i^*)



2. Methodology & Analytical Steps

The TOPSIS model standardizes all evaluation metrics through a 6-step algorithmic sequence:

- **Step 1 (Matrix Construction):** Formulated baseline scores for 6 ERP alternatives against 9 Criteria (3 Cost, 6 Benefit).
- **Step 2 & 3 (Vector Normalization & Weighting):** Applied Euclidean normalization ($r_{ij} = x_{ij} / \sqrt{\sum x_{ij}^2}$) and criteria weight multipliers (w_j).
- **Step 4 & 5 (Ideal Separation Measures):** Calculated Euclidean distances (S_i^*, S_i^-) from Positive Ideal (A^*) and Negative Ideal (A^-) solutions.
- **Step 6 (Relative Closeness):** Derived closeness kpis (C_i^*) to produce the unambiguous final selection ranking.

3. Strategic Recommendation

- **Primary Recommendation (ERP 5):** Ranked 1st ($C_5^* = 0.6558$) with the shortest distance to the ideal solution ($S_5^* = 0.0225$). It provides maximum utility across operational benefit criteria.
- **Alternative Choice (ERP 6):** Ranked 2nd ($C_6^* = 0.5690$), standing out as a resilient fallback option.
- **Vendor Rejection (ERP 3):** Lowest scoring platform ($C_3^* = 0.3539$) due to highest integration downtime and lowest relative